# Lecture 5: Inferences About a Mean Vector

Weihua Zhou

February 15, 2022

# Introduction

This lesson deals with inference problems on multivariate population mean. Similar to univariate population, inference problems may arise in three different scenarios:

- Inference problems regarding a single multivariate population
- Inference problems regarding two means from a paired population
- Inference problems regarding two means from two independent populations

In this lesson we will focus on Case I and defer other two cases to future class.

# Lesson Objectives

Upon completion of this lesson, you should be able to do the following:

- What is the distribution of $\bar{X}$ when data come from a multivariate normal distribution?
- What are the properties of $\bar{X}$ when data do NOT come from a multivariate normal distribution but the sample size n is large?
- Use SAS to carry out Hotelling's T-square test for the null hypothesis that a population mean vector meets specifications;
- Compute and interpret simultaneous and Bonferroni confidence intervals;

# Distribution of Sample Mean Vector

As noted previously $\bar{X}$ is a function of the random data, and hence $\bar{X}$ is also a random vector with a mean, a variance-covariance matrix and a distribution.

- **Mean:** We have already seen that the mean of the sample mean vector is equal to the population mean vector $\mu$:

$$E(\bar{X}) = E(X) = \mu.$$

- **Variance:**
  1. *Univariate Setting*:

$$var(\bar{x}) = \frac{\sigma^2}{n}$$

This, of course, is a function of the unknown population variance $\sigma^2$. We can estimate this by simply substituting $s^2$ in the sample variance $\sigma^2$ yielding our estimate for the variance of the population mean as shown below:

$$\widehat{var}(\bar{x}) = \frac{s^2}{n}$$

# Variance of Sample Mean Vector

*Multivariate Setting*: The population variance-covariance matrix replaces the variance of the $\bar{x}$'s generated from independent samples of size $n$, taking a form similar to the univariate setting as shown below:

$$var(\bar{X}) = \frac{1}{n}\Sigma$$

Again, this is a function of the unknown population variance-covariance matrix $\Sigma$. An estimate of the variance-covariance matrix of $\Sigma$ can be obtain by substituting the sample variance-covariance matrix $S$ for the population variance-covariance matrix $\Sigma$, yielding the estimate as shown below:

$$var(\bar{X}) = \frac{1}{n}S$$

# Distribution

Let's consider the distribution of the sample mean vector, first looking at the univariate setting and comparing this to the multivariate setting.

- *Univariate Setting*: Here we are going to make the additional assumption that $X_1, X_2, \ldots, X_n$ are independently sampled from a normal distribution with mean $\mu$ and variance $\sigma^2$. In this case, $\bar{x}$ is normally distributed as

$$\bar{x} \sim N(\mu, \frac{\sigma^2}{n})$$

- *Multivariate Setting*: Similarly, for the multivariate setting, we are going to assume that the data vectors $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ are independently sampled from a multivariate normal distribution with mean vector $\mu$ and variance-covariance matrix $\Sigma$. Then, in this case, one has

$$\bar{\mathbf{X}} \sim N(\mu, \frac{1}{n}\Sigma)$$

# Law of Large Numbers

At this point we will drop the assumption that the individual observations are sampled from a normal distribution and look at laws of large numbers. These will hold regardless of the distribution of the individual observations.

- *Univariate Setting*: In the univariate setting, we see that if the data are independently sampled, then the sample mean, $\bar{X}$, is going to converge (in probability) to the population mean $\mu$. What does this mean exactly? It means that as the sample size gets larger and larger the sample mean will tend to approach the true value for a population $\mu$.
- *Multivariate Setting*: A similar result is involved in the multivariate setting , the sample mean vector, $\bar{\mathbf{X}}$, will also converge (in probability) to the mean vector $\mu$; As our sample size gets larger and larger, each of the individual components of that vector, $\bar{X}_j$, will converge to the corresponding mean, $\mu_j$.

$$\bar{X}_j \xrightarrow{p} \mu_j$$

# Central Limit Theorem

Just as in the univariate setting we also have a multivariate Central Limit Theorem. But first, let's review the univariate Central Limit Theorem.

- *Univariate Setting*. If all of our individual observations, $X_1, X_2, \ldots, X_n$, are independently sampled from a population with mean $\mu$ and variance $\sigma^2$, then, the sample mean, $\bar{X}$, is approximately normally distributed with mean $\mu$ and variance $\sigma^2/n$. Note the Central Limit Theorem is a general result which holds regardless of the distribution of the original data. Significance of CLT lies in the fact that the sample mean is approximately normally distributed for large samples whatever be the distribution of the individual observations.
- *Multivariate Setting*. A similar result is available in the multivariate setting. If our data vectors $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$, are independently sampled from a population with mean vector $\mu$ and variance-covariance matrix $\Sigma$, then the sample mean vector, $\bar{\mathbf{X}}$, is going to be approximately normally distributed with mean vector $\mu$ and variance-covariance matrix $\frac{1}{n}\Sigma$.

# Motivation

In 1985, the USDA commissioned a study of women's nutrition. Nutrient intake was measured for a random sample of 737 women aged 25-50 years. Five nutritional components were measured: calcium, iron, protein, vitamin A and vitamin C. In previous analyses of these data, the sample mean vector was calculated. The table below shows the recommended daily intake and the sample means for all the variables:

| **Variable** | **Recommended Intake** $\mu_0$ | **Mean** |
|---|---|---|
| Calcium | 1000 mg | 624.0 mg |
| Iron | 15mg | 11.1 mg |
| Protein | 60g | 65.8 g |
| Vitamin A | 800$\mu$g | 839.6 $\mu$g |
| Vitamin C | 75 mg | 78.9 mg |

One of the questions of interest is whether women meet the federal nutritional intake guidelines. If they fail to meet the guidelines, then we might ask for which nutrients the women fail to meet the guidelines.

# Motivation

The hypothesis of interest is that women meet nutritional standards for all nutritional components. This null hypothesis would be rejected if women fail to meet nutritional standards on any one or more of these nutritional variables. In mathematical notation, the null hypothesis is the population mean vector $\mu$ equals the hypothesized mean vector $\mu_0$ as shown below:

$$H_0 : \mu = \mu_0$$

Let us first compare the univariate case with the analogous multivariate case in the following tables.

**Focus of Analysis**

| **Univariate Case** | **Multivariate Case** |
|---|---|
| Measuring only a single nutritional component Data: scalar quantities $X_1, X_2, \ldots, X_n$ | Data: $p \times 1$ random vectors: $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ |

# Univariate vs. Multivariate

**Assumptions Made In Each Case**

| **Univariate Case** | **Multivariate Case** |
|---|---|
| **1. Mean** | |
| Common mean $\mu$; $E(X_i) = \mu, \quad i = 1, 2, \ldots, n$ | Common mean vector $\mu$; $E(\mathbf{X}_i) = \mu, \quad i = 1, 2, \ldots, n$ |
| **2. Homoskedasticity** | |
| Common variance $\sigma^2$; $var(X_i) = \sigma^2, \quad i = 1, 2, \ldots, n$ | Common variance-covariance matrix $\Sigma$: $var(\mathbf{X}_i) = \Sigma, \quad i = 1, 2, \ldots, n$ |
| **3. Independence** | |
| The subjects are independently sampled. | The subjects are independently sampled. |
| **4. Normality** | |
| The subjects are sampled from a normal distribution | The subjects are sampled from a multivariate normal distribution. |

# Univariate vs. Multivariate

**Hypothesis Testing in Each Case**

| **Univariate Case** | **Multivariate Case** |
|---|---|
| Consider hypothesis testing: $H_0 : \mu = \mu_0$ against alternative $H_a : \mu \neq \mu_0$ | Consider hypothesis testing: $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ against $H_0 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$ We can also write this expression as shown below: $H_0 : \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} = \begin{pmatrix} \mu_1^0 \\ \mu_2^0 \\ \vdots \\ \mu_p^0 \end{pmatrix} \quad vs \quad H_a : \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} \neq \begin{pmatrix} \mu_1^0 \\ \mu_2^0 \\ \vdots \\ \mu_p^0 \end{pmatrix}$ |

# T-test for Univariate

In your introductory statistics course you would have learned that such a null a hypothesis can be tested using a $t$-statistic as shown in the expression below:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

Under $H_o$ this $t$-statistic is going to have a $t$ distribution with $n-1$ degrees of freedom. We would reject $H_o$ at level $\alpha$ if the absolute value of the test statistic $t$ is greater than the critical value from the $t$-table, evaluated at $\alpha/2$ as shown below:

$$|t| > t_{\alpha/2,n-1}$$

# A Naive Approach

Following univariate method, a naive approach for testing multivariate hypothesis is to compute the $t$-test statistics for each individual variable; i.e.,

$$t_j = \frac{\bar{x}_j - \mu_j^0}{s_j/\sqrt{n}} \sim t_{n-1}, \quad j = 1, \ldots, p$$

Thus we could define $t_j$, which would be the $t$-statistic for the $j$th variable as shown above. We may then

$$\text{reject } H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0 \ \text{ if } \ |t_j| > t_{n-1,\alpha/2}$$

for at least one variable $j \in \{1, 2, \ldots, p\}$ .

# Problem with Family-wide error rate

The basic problem with this naive approach is that it does not control for family wide error rate. By definition, the family-wide error rate is the probability of rejecting at least one of the hypothesis tests $H_0^{(j)} : \mu_j = \mu_j^0$ when all of the $H_0$'s are true.

To understand the family wide error rate suppose that the experimental variance-covariance matrix to be diagonal. This implies zero covariances between the variables. If the data are multivariate normally distributed then this would mean that all of the variables are independently distributed. In this case, the family wide error rate is

$$1 - (1 - \alpha)^p > \alpha$$

where p is the dimension of the multivariate date and $\alpha$ is the level of significance. By definition, the family wide error rate is equal to the probability that we reject $H_0^{(j)}$ for at least one $j$, given that $H_0^{(j)}$ is true for all $j$. Unless $p$ is equal to one, the error will strictly be greater than $\alpha$.

# Bonferroni Correction

**Consequence**

The naive approach yields a liberal test. That is, we will tend to reject the null hypothesis more often than we should.

**Bonferroni Correction**

Let $H_0^{(1)}, ..., H_0^{(m)}$ be a family of hypotheses and $P_1, ..., P_m$ the corresponding P-values. The Bonferroni Correction states that rejecting all $H_0^{(j)}$ via $P_j < \frac{\alpha}{m}$ will control the $FWER \leq \alpha$. The proof follows from Boole's inequality:

$$\begin{aligned} FWER &= P(\text{reject at leat one } H_0^{(j)} | H_0 \text{ is true}) \\ &= P(\cup\{\text{reject } H_0^{(j)}\} | H_0 \text{ is true}) \\ &\leq \sum_{j=1}^{m} P(\text{reject } H_0^{(j)} | H_0 \text{ is true}) \\ &= m \times \frac{\alpha}{m} = \alpha \end{aligned}$$

The equality will hold when the tests are independent.

# Bonferroni Correction

Under the Bonferroni Correction we would reject the null hypothesis that mean vector $\boldsymbol{\mu}$ is equal to our hypothesized mean vector $\boldsymbol{\mu}_0$ at level $\alpha$, if, for at least one variable $j$, the absolute value of $t_j$ is greater than the critical value from the $t$-table with $n-1$ degrees of freedom evaluated at $\alpha/(2p)$ for at least one variable $j$ between 1 and $p$.

$$\text{reject} \quad H_0 \quad \textit{if} \quad |t_j| > t_{n-1,\alpha/(2p)} \quad \text{for some } j$$

The problem with the Bonferroni Correction, however, is that it assumes independence among the variables. In a multivariate setting, the assumption of independence between variables is likely to be violated.

**Consequence**

For correlated variables, the true family wide error rate lies somewhere between $\alpha$ and $\alpha/p$. The test may be too conservative, and so, there will be low power to detect the desired effects.

# Hotelling's T-Square

As an alternative test statistic the Hotelling's $T^2$ is going to be used.
To motivate Hotelling's $T^2$, consider the square of the $t$-statistic for testing hypothesis regarding univariate mean:

$$t^2 = \frac{(\bar{x} - \mu_0)^2}{s^2/n} = n(\bar{x} - \mu_0)(s^2)^{-1}(\bar{x} - \mu_0) \sim F_{1,n-1}.$$

If you square a $t$-distributed random variable with $n-1$ degrees of freedom you get an $F$-distributed random variable with 1 and $n-1$ degrees of freedom. Now we could reject $H_0$ at level $\alpha$ if $t^2$ is greater than the critical value from the $F$-table with 1 and $n-1$ degrees of freedom, evaluated at level $\alpha$:

$$t^2 > F_{1,n-1}(\alpha)$$

# Hotelling's T-Square

Consider the last term in the above expression for $t^2$. In the expression for Hotelling's $T^2$, the difference between the sample mean and the $\mu_0$ is replaced with the difference between the sample mean vector and the hypothesized mean vector $\boldsymbol{\mu}_0$. The sample variance is replaced by the inverse of the sample variance-covariance matrix $\mathbf{S}$ yielding the expression below:

$$T^2 = n(\bar{\boldsymbol{x}} - \boldsymbol{\mu}_0)'\boldsymbol{S}^{-1}(\bar{\boldsymbol{x}} - \boldsymbol{\mu}_0)$$

- Under the multivariate normal assumption,

$$T^2 \sim \frac{(n-1)p}{(n-p)} F_{p,n-p}$$

- For large $n$, $T^2$ is approximately chi-square distributed with $p$ degrees of freedom.

# Hotelling's T-Square

Since

$$T^2 \sim \frac{(n-1)p}{(n-p)} F_{p,n-p}$$

we can compute $T^2$ and compare it to

$$\frac{(n-1)p}{(n-p)} F_{p,n-p}(\alpha)$$

Or use the fact that

$$\frac{(n-p)}{(n-1)p} T^2 \sim F_{p,\ (n-p)}$$

and find the P-value

$$P - value = Prob\left\{ F_{p,\ (n-p)} \geq \frac{(n-p)}{(n-1)p} T^2 \right\}$$

Reject $H_0$ when the P-value is small (i.e., $T^2$ is large).

- Why $F$-test is always right tailed here?

# After Rejection: Confidence Regions

Our goal is to make inferences about populations from samples. In univariate statistics, we form confidence intervals; we'll generalization this to multivariate confidence region.
General definition: A confidence region is a region of likely values of parameters $\theta$ which is determined by data:

$$R(\boldsymbol{X}) = \text{confidence region}$$

where

- $\boldsymbol{X}' = (\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n)$; that is, data.
- $R(\boldsymbol{X})$ is a $100(1-\alpha)\%$ confidence region if before the sample was selected.

$$Prob[R(\boldsymbol{X})\text{contains the true } \theta] = 1 - \alpha$$

# Confidence Region for $\boldsymbol{\mu}$

For a typical sample,

1. Calculate $\bar{\boldsymbol{x}}$ and $S$.
2. Find $(n-1)p/(n-p)F_{p,n-p}(\alpha)$.
3. Consider all $\boldsymbol{\mu}$'s that satisfy the equation

$$\boxed{n(\bar{\boldsymbol{x}}-\boldsymbol{\mu})'\boldsymbol{S}^{-1}(\bar{\boldsymbol{x}}-\boldsymbol{\mu}) \leq \frac{(n-1)p}{(n-p)}F_{p,n-p}(\alpha)}$$

This is the confidence region, which is an equation of an ellipsoid.

# Alternatives to Confidence Regions

The confidence regions consider all the components of $\boldsymbol{\mu}$ jointly.

We often desire a confidence statement (i.e, confidence interval) about individual components of $\boldsymbol{\mu}$ or a linear combination of the $\mu_j$'s.

We want all such statements to hold simultaneously with some specified large probability; that is, we want to make sure that the probability that any one of the confidence statements is incorrect is small.

# General Approach

The **general format** of a confidence interval estimate of a population mean is

$$\text{Sample mean} \pm \text{Multiplier} \times \text{Standard error of mean}$$

For variable $X_j$, a confidence interval estimate of its population mean $\mu_j$ is

$$\bar{x}_j \pm \text{Multipliers} \frac{s_j}{\sqrt{n}}$$

In this formula, $\bar{x}_j$ is the sample mean, $s_j$ is the sample standard deviation and $n$ is the sample size. The multiplier value is a function of the confidence level, the sample size, and the strategy used for dealing with the multiple inference issue.

# Strategies for Determining the Multiplier

Some common strategies are

- One at a time confidence intervals: This strategy essentially considers each component separately and uses the desired confidence level (usually 95%) for each single interval.
- Simultaneous Confidence Intervals: This strategy uses properties of the multivariate normal distribution to define joint confidence intervals. The multiplier for this method is conservative because the family error rate applies to the family of all possible linear combinations of population means.
- Bonferroni Method: With this method we set a family wide error rate and then divide this family error rate by the number of intervals to be computed to determine the error rate (and hence confidence level) for each individual interval.

# One at a time confidence intervals

For a $1 - \alpha$ confidence interval, the "one at a time" multiplier is the $t$-value such that the probability is $1 - \alpha$ between $t$ and $+t$ under a $t$-distribution with $n - 1$ degrees of freedom. Said another way, the value of $t$ is such that the probability greater than $+t$ is $\alpha/2$. Notationally

$$\text{Multiplier} = t_{n-1}(\alpha/2).$$

With this notation a confidence interval for $\mu_j$ is computed as

$$\bar{x}_j \pm t_{n-1}(\alpha/2)\frac{s_j}{\sqrt{n}}$$

# Drawback of One-At-A-Time Method

When we determine confidence intervals for the population means of several variables, we are creating a family of confidence intervals. The **family-wide error rate** is the probability that at least one of the confidence intervals in the family will not capture the population mean. The **family-wide confidence level**$= 1-$ family-wide error.

Problem with One-At-A-Time Method: If the confidence level is $100(1-\alpha)\%$ for one interval, then the **family-wide confidence level** could be much less that $100(1-\alpha)\%$.

Example: Suppose we have $p$ confidence intervals for $\mu_1, \mu_2, \ldots, \mu_p$. Each confidence interval has confidence level $100(1-\alpha)\%$. Assume $X_1, \ldots, X_p$ are independent. What is the family-wide confidence level for this group of confidence intervals?

# Simultaneous Confidence Intervals ($T^2$ intervals)

This method is derived from properties of the multivariate normal distribution. The multiplier applies to the family of all possible linear combinations of the population means being considered, including the individual means. It is conservative (large value). Consider linear combinations of population means as below:

$$\boldsymbol{a}'\boldsymbol{\mu} = a_1\mu_1 + a_2\mu_2 + \ldots + a_p\mu_p$$

Let $X_1, X_2, \ldots, X_n$ be a random sample from $N_p(\boldsymbol{\mu}, \Sigma)$ population with $det(\Sigma) > 0$, then simultaneously for all $\boldsymbol{a}$, the interval

$$\boldsymbol{a}'\bar{\boldsymbol{x}} \pm \sqrt{\frac{p(n-1)}{n-p} F_{p,n-p}(\alpha)} \sqrt{\frac{\boldsymbol{a}' S \boldsymbol{a}}{n}}$$

will contain $\boldsymbol{a}'\boldsymbol{\mu}$ with coverage rate $100(1-\alpha)\%$.

In terms of interpreting the $100(1-\alpha)\%$ confidence ellipse, we can say that we are $100(1-\alpha)\%$ confident that all such confidence intervals cover their respective linear combinations of the treatment means, regardless of what linear combinations we may

# Simultaneous Confidence Intervals ($T^2$ intervals)

In particular, we can consider the trivial linear combinations which correspond to the individual variables. To get an interval for a each of the $p$ means, set

$$a_i' = (0, 0, \ldots, \underset{ith}{1}, 0, \ldots, 0), i = 1, \ldots, p.$$

Therefore, the simultaneous confidence intervals are

$$\bar{x}_j \pm \sqrt{\frac{p(n-1)}{n-p} F_{p,n-p}(\alpha)} \frac{s_j}{\sqrt{n}}$$

# Comparison

| One-at-a-Time | $T^2$ Intervals |
|---|---|
| ▶ Narrower (more precise) | ▶ Wider (less precise) |
| ▶ More powerful | ▶ Less powerful |
| ▶ Liberal | ▶ Conservative |
| ▶ Coverage rate $< 100(1-\alpha)\%$ | ▶ Coverage rate $\geq 100(1-\alpha)\%$ |
| ▶ Coverage rate depends on number of intervals and $S$. | ▶ Coverage rate does not depend on number of intervals. |
| ▶ Accuracy may be OK provided that first test (& reject) $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$. | ▶ Good if do a lot of intervals, (e.g., $> p$) |

Compromise: Bonferroni

# Bonferroni Intervals

The problem with the simultaneous confidence intervals is that if we are not interested in all possible linear combinations of variables or anything other than the means by themselves, then the simultaneous confidence interval may be too wide, and hence, too conservative.

As an alternative to the simultaneous confidence intervals we can use the Bonferroni intervals. This method will

- Give narrower (more precise) intervals than $T^2$, but not as narrow are the univariate ones.
- Good if
  1. The intervals that you construct are decided upon a *priori*.
  2. You only construct $\leq p$ intervals.
- It uses a form of the Bonferroni inequality.

# Bonferroni Inequality

Given $m$ intervals,

$$\begin{aligned}
\text{Prob}\{\text{all intervals are valid}\} &= 1 - \text{Prob}\{\text{at least 1 false}\} \\
&\geq 1 - \sum_{i=1}^{m} \text{Prob}\{i\text{th interval is false}\} \\
&= 1 - \sum_{i=1}^{m} (1 - \text{Prob}\{i\text{th interval is true}\}) \\
&= 1 - \sum_{i=1}^{m} \alpha_i
\end{aligned}$$

This is a form of the Bonferroni inequality:

$$P\{\text{all intervals are ture}\} \geq 1 - (\alpha_1 + \alpha_2 + \ldots + \alpha_m)$$

We set $\alpha_i = \alpha/m$ using a pre-determined $\alpha$-level, then

$$P\{\text{all intervals are ture}\} \geq 1 - m \times \alpha/m = 1 - \alpha.$$

# Bonferroni Intervals

Suppose that we are calculating $p$ intervals with a family error rate equal to $\alpha$. Notationally, the Bonferroni method multiplier is

$$\text{Multiplier} = t_{n-1}(\alpha/2p).$$

A confidence interval for $\mu_j$ is computed as

$$\bar{x}_j \pm t_{n-1}(\alpha/2p)\frac{s_j}{\sqrt{n}}$$

# Example: Womens Nutrition Data and Associated Confidence Intervals

It turns out that SAS does not have any procedures for calculating Hotelling's $T^2$. So in this case we are going to have to rely on the iml procedure, which can carry out matrix manipulation. And in this case, it can be used for carrying out Hotelling's $T^2$ test.

The code is provided in nutrient4.txt. To use this program, I recommend copying my program which I have given here and then just making the necessary changes to use it with your dataset on your homework assignment. Again, only three entries are all we really need to change to use this with any dataset. First the specified value of $\boldsymbol{\mu}_0$. Second, which dataset you want to use, and third, the specification of what variable you want to analyze.

# Example

The recommended intake and sample mean are given below

| **Variable** | **Recommended Intake** $\mu_0$ | **Mean** |
|---|---|---|
| Calcium | 1000 mg | 624.0 mg |
| Iron | 15mg | 11.1 mg |
| Protein | 60g | 65.8 g |
| Vitamin A | 800$\mu$g | 839.6 $\mu$g |
| Vitamin C | 75 mg | 78.9 mg |

as well as the sample variance-covariance matrix:

$$S = \begin{pmatrix} 157829.4 & 940.1 & 6075.8 & 102411.1 & 6701.6 \\ 940.1 & 35.8 & 114.1 & 2383.2 & 137.7 \\ 6075.8 & 114.1 & 934.9 & 7330.1 & 477.2 \\ 102411.1 & 2383.2 & 7330.1 & 2668452.4 & 22063.3 \\ 6701.6 & 137.7 & 477.2 & 22063.3 & 5416.264 \end{pmatrix}$$

# Example

Hotelling's T-square comes out to be:

$$T^2 = 1758.5413$$

The F-statistic is:

$$F = 349.80 > 3.02 = F_{5,732}(0.01)$$

For an 0.01 level test, the critical value is approximately 3.02. Since 349.80 is greater that this value, we can reject the null hypothesis that the average woman meets dietary intake recommendations.

$$(T^2 = 1758.54; F = 349.80; d.f. = 5, 732; p < 0.0001)$$

The SAS program computes the p-value to be about 0.00. p-values actually are never equal to zero so we can just simply state that the p-value is less than 0.0001.

# Example

**Conclusion**

The average woman between 25 and 50years old does not meet recommended dietary intake standards. Returning to the table of sample means and recommended dietary intake, it appears that women fail to meet nutritional standards for calcium and iron, and perhaps exceed intakes for protein, vitamin A and vitamin C.

**A Question Emerges...**

For which nutrients do women fall significantly below recommended nutritional intake levels? Or, conversely, for what nutrients do women fall significantly above recommended nutritional intake levels?

# One-at-a-time intervals

The one-at-a-time intervals, using univariate method as shown below:

$$\bar{x}_j \pm t_{n-1}(\alpha/2)\sqrt{s_j^2/n}$$

If we consider only a single variable, we can say with $(1-\alpha)\times 100\%$ confidence that the interval includes the corresponding population mean. Note that $t_{736,0.025} = 1.96$. The one-at-a-time confidence intervals are summarized in the table below:

| **Variable** | $\mu_0$ | 95% Confidence Interval |
|---|---|---|
| Calcium | 1000 mg | (595.3, 652.8) |
| Iron | 15mg | (10.7, 11.6) |
| Protein | 60g | (63.6, 68.0) |
| Vitamin A | 800$\mu$g | (721.5, 957.8) |
| Vitamin C | 75 mg | (73.6, 84.2) |

# One-at-a-time intervals

Looking at this table, it appears that the average woman falls below recommended daily intakes of calcium and iron (since the intervals fall below the recommended intakes of these variables), and exceeds the recommended daily intake of protein (since the interval falls above the recommended intake of protein).

Problem: The problem with these one-at-a-time intervals is that they do not control for family-wide error rate.

Consequence: We are less than 95% confident that all of the intervals simultaneously cover their respective means.

# Simultaneous confidence intervals

Simultaneous confidence intervals are computed using hand calculations. Here we have $p = 5$ variables, a total sample size of $n = 737$, and if we look up in the F-table for 5 and 732 degrees of freedom for $\alpha = 0.05$, the critical value is 2.21. Thus, for *calcium*, one has

$$\begin{aligned} 624.05 \quad &\pm \quad \sqrt{\frac{5(737-1)}{737-5} F_{5,737-5}(0.05)} \sqrt{157829.4/737} \\ &= \quad (575.27, 672.83) \end{aligned}$$

The following table gives the $T^2$ confidence intervals:

| **Variable** | $\mu_0$ | 95% Confidence Interval |
|---|---|---|
| Calcium | 1000 mg | (575.1, 673.0) |
| Iron | 15mg | (10.4, 11.9) |
| Protein | 60g | (62.0, 69.6) |
| Vitamin A | 800$\mu$g | (638.3, 1040.9) |
| Vitamin C | 75 mg | (69.9, 88.0) |

# Simultaneous confidence intervals

Looking at these simultaneous confidence intervals we can see that the upper bound of the interval for calcium falls below the recommended daily intake of 1000 mg. Similarly, the upper bound for iron also falls below the recommended daily intake of 15 mg. Conversely, the lower bound for protein falls above the recommended daily intake of 60 g for protein. The intervals for both vitamin A and C both contain the recommended daily intake for these two vitamins.

Therefore, we can conclude that the average woman falls below the recommended daily intakes of calcium and iron, and exceeds the recommended daily intake of protein.

The problem with the simultaneous confidence intervals is that it may be too wide, and hence, too conservative. As an alternative to the simultaneous confidence intervals we can use the Bonferroni intervals.

# Bonferroni intervals

The Bonferroni intervals as given in the expression below:

$$\bar{x}_j \pm t_{n-1}(\alpha/2p)\sqrt{s_j^2/n}$$

Here $n = 737$, $\alpha = .05$ and $p = 5$. Therefore,

$$t_{737-1}(0.05/10) = 2.576$$

For example, the 95% confidence interval for *calcium* under the Bonferroni correction is calculated as below:

$$\begin{aligned} 624.05 &\pm 2.576 \times \sqrt{157829.4/737} \\ &= (586.35, 661.75) \end{aligned}$$

# Bonferroni intervals

The table below shows the results of the computation

| **Variable** | $\mu_0$ | 95% Confidence Interval |
|---|---|---|
| Calcium | 1000 mg | (586.3, 661.8) |
| Iron | 15mg | (10.6, 11.7) |
| Protein | 60g | (62.9, 68.7) |
| Vitamin A | 800$\mu$g | (684.2, 995.0) |
| Vitamin C | 75 mg | (71.9, 85.9) |

When compared to the simultaneous intervals, we see that the Bonferroni intervals are narrower. However, in this case, the conclusions will be the same. The confidence intervals for both vitamin A and C both cover their respective recommended daily intakes. Intervals for calcium and iron fall below the recommended intake, while the interval for protein falls above it.

This suggests that the average woman falls below recommended daily intakes of calcium and iron, and exceeds recommended daily intake of protein.

# SAS code for confidence intervals

The SAS code nutrient5.txt can be used to generate confidence intervals for population means using the three methods discussed here. The code reads a dataset, reshapes it to have a data line for each variable value, determines means and standard deviations and then calculates and prints the three types of intervals. To use this code for different situations, you need only to change the third line where the value of $p$ is set and the data step where the data set is read and reshaped.